

W-DRO-T Training Procedure

Stage 1: CVaR Pre-train

50 epochs

$$\mathcal{L} = \text{CVaR}_{0.95}$$

LR= 10^{-3}

Stage 2: Entropic + DRO

30 epochs

$$\mathcal{L} = \rho_{\lambda} + \varepsilon \|\nabla\|_2$$

$\varepsilon : 0 \rightarrow 0.1$

ε -Annealing Schedule

Linear warmup 15ep

SDPA Math backend

create_graph=True

Key Implementation Detail:

DRO gradient penalty needs

create_graph=True for ∇^2

→ forces SDPA Math backend

(no Flash/Efficient Attention)

Robustness Guarantee (Blanchet & Murthy 2019):

$$\begin{aligned} & \sup_{Q: W(P, Q) \leq \varepsilon} E_Q[\ell] \\ & \approx E_P[\ell] + \varepsilon \cdot E_P[\|\nabla \ell\|_2] \end{aligned}$$

First-order approximation of worst-case risk